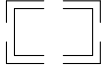
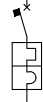
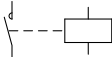
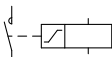
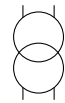


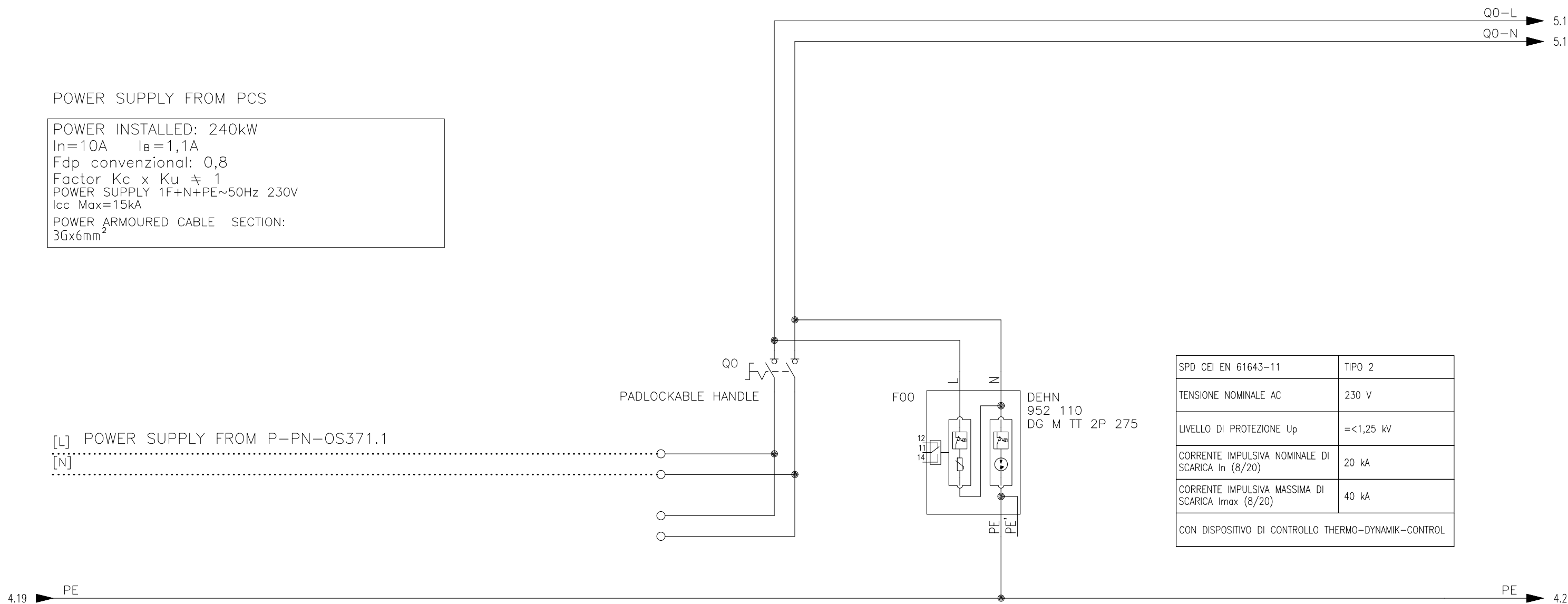
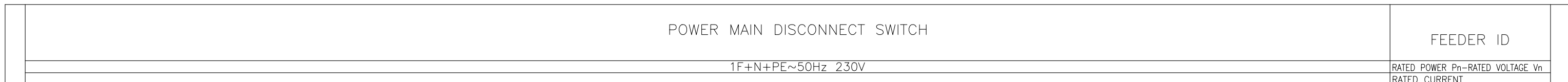
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TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																					
SAFETY CRITICAL ELEMENT LIST																					
IFI	1	09/12/2025	AS-BUILT				ADF	GDP	IC												
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC												
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.												
Rev. Index		Date	Description				Prepared	Checked	Approved												
Client							Project Title														
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT														
Contractor							Provision of a m-LNG plant for production for ManaEnergy														
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																					
Engineering																					
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																					
Facility Name			Location				Scale		Sheet of Sheets												
TENDRARA			MOROCCO				A4		1 of 16												
Document Title					Doc. Code					esa esaelettromeccanica.com INGEGNERIA ELETTRICA - QUADRI DI DISTRIBUZIONE E AUTOMAZIONE - CABINE DI TRASFORMAZIONE MT/BT – PLC – HMI – SCADA - ENERGIE RINNOVABILI CONTRADA TESINO, 47 - 63065 RIPATRANSONE (AP) – ITALY - TEL. +39 0735 90523 WWW.ESAELETTROMECCANICA.COM - INFO@ESAELETTROMECCANICA.COM - ESA@PEC.IT - P.IVA e C.F. 00768110447											
P-PN-0S354.1					ELD01-ING-FR01																
AMMONIA MR PRECCOLER SKID					Vendor Doc. code																
IE24090_SE10_P-PN-OS354.1																					
			Project :				Customer :				Object :										
			P–PN–0S354.1				ManaEnergy Managem Group								By :						
General Contractor :		Manufacture :		AMMONIA MR PRECCOLER SKID				Contract :				Rev.		Data :				Date :			
		esa esaelettromeccanica.com						S/N :				Rev.		Data :				Date :		Sheet :	
								IE24090				1		09/12/2025				26/03/2024		1 of 16	
								IE24090_SE10													

CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

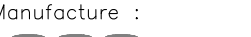
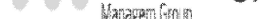
CONDUCTOR COLOR CODING INSIDE CONTROL PANEL	
AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT–BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW–GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK–BLUE
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED	

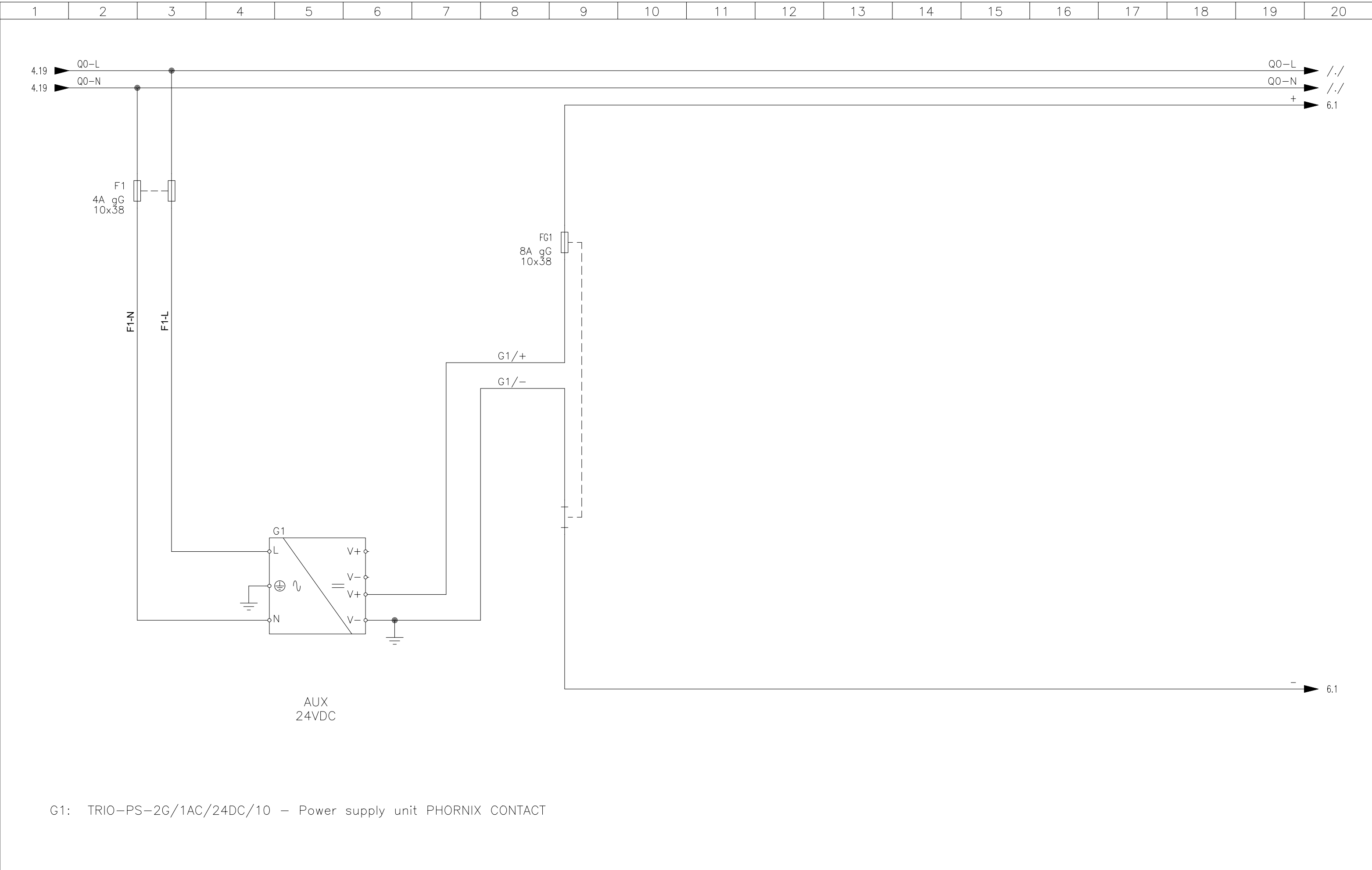
CODE COLOUR	
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE



SPD CEI EN 61643-11	TIPO 2
TENSIONE NOMINALE AC	230 V
LIVELLO DI PROTEZIONE Up	=<1,25 kV
CORRENTE IMPULSIVA NOMINALE DI SCARICA In (8/20)	20 kA
CORRENTE IMPULSIVA MASSIMA DI SCARICA I _{max} (8/20)	40 kA
CON DISPOSITIVO DI CONTROLLO THERMO-DYNAMIK-CONTROL	

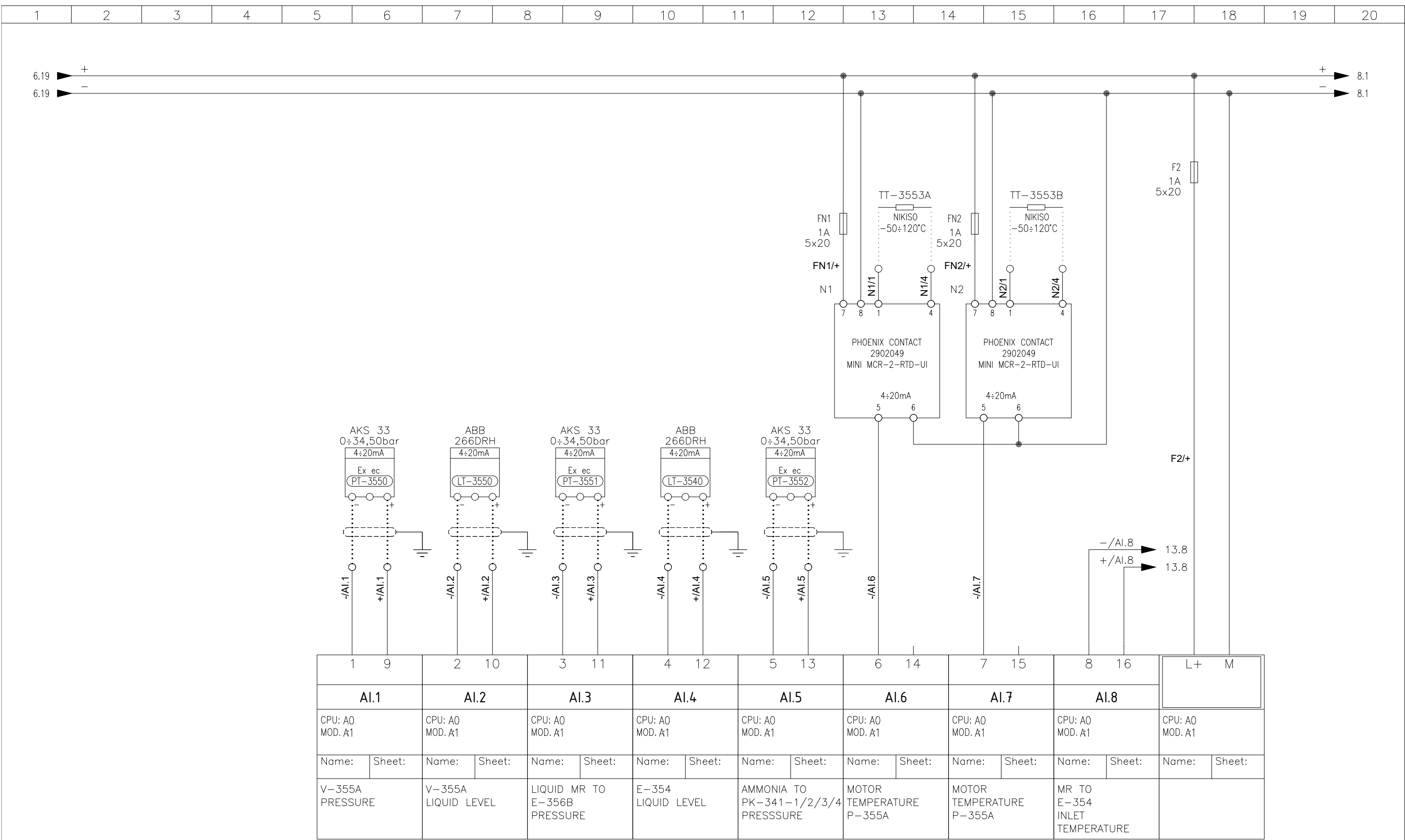
CONDUCTOR	PROTECTION		TYPE	CONDUCTOR	PROTECTION
			RATED BREAKING CAPACITY Icu		
			RATED or THERMAL CURRENT In or Ith		
			MAGNETIC RELEASE CHARACTERISTICS Im		
			RATED R.C. SENSITIVITY Idn		
CONDUCTOR	PROTECTION		WIRE SECTION	CONDUCTOR	PROTECTION
			CABLE TYPE		
			CABLE CARRYING LOAD Iz		
			CABLE LENGHT		
			LINE ID		
		WIRE ID ON TERMINAL			

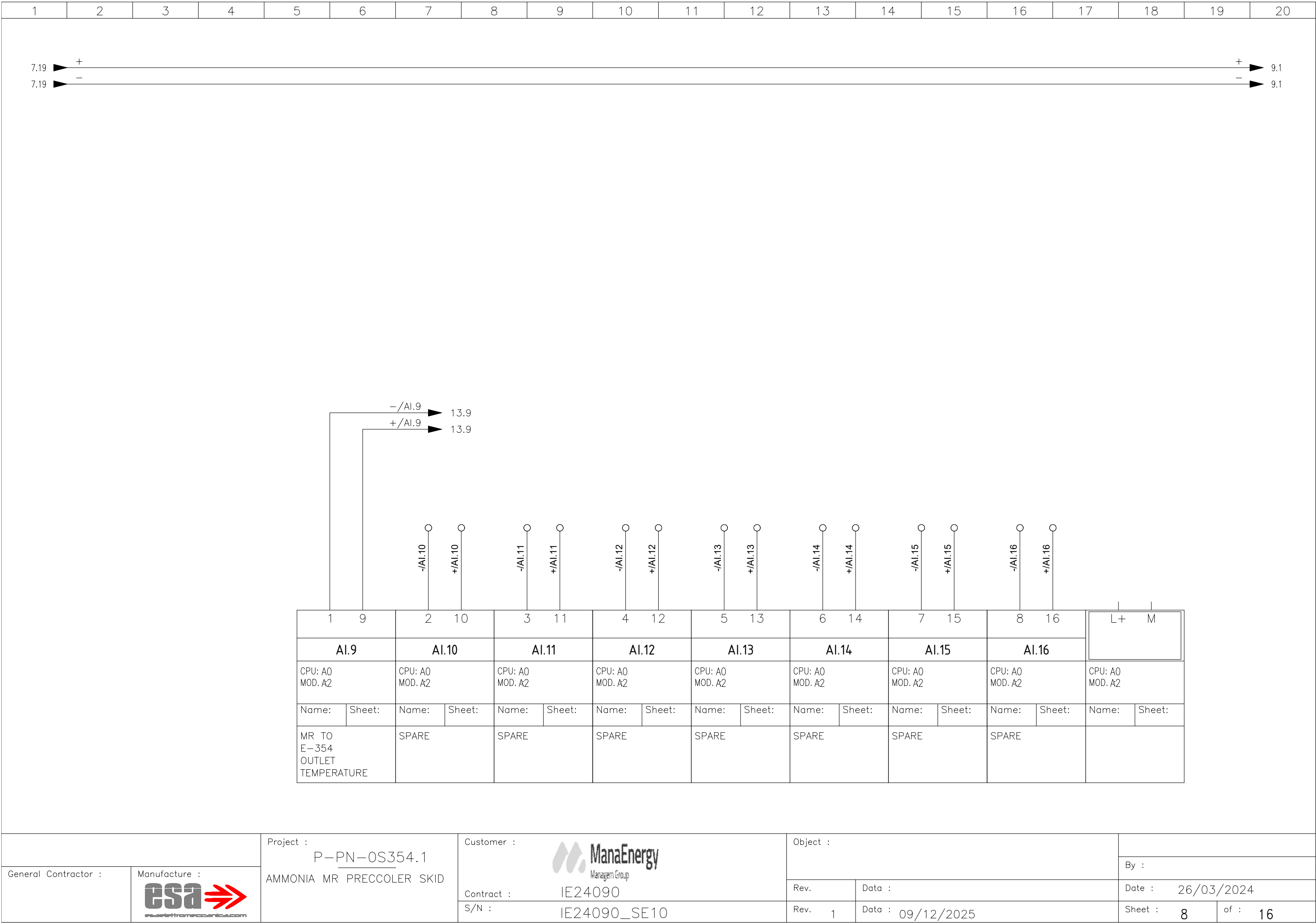
General Contractor :		Manufacture : 	Project : P-PN-0S354.1 AMMONIA MR PRECCOLER SKID	Customer :  Management Group	Object :			
					By :			
				Contract : IE24090	Rev.	Data :	Date : 26/03/2024	
				S/N : IE24090_SE10	Rev. 1	Data : 09/12/2025	Sheet : 4	of : 16

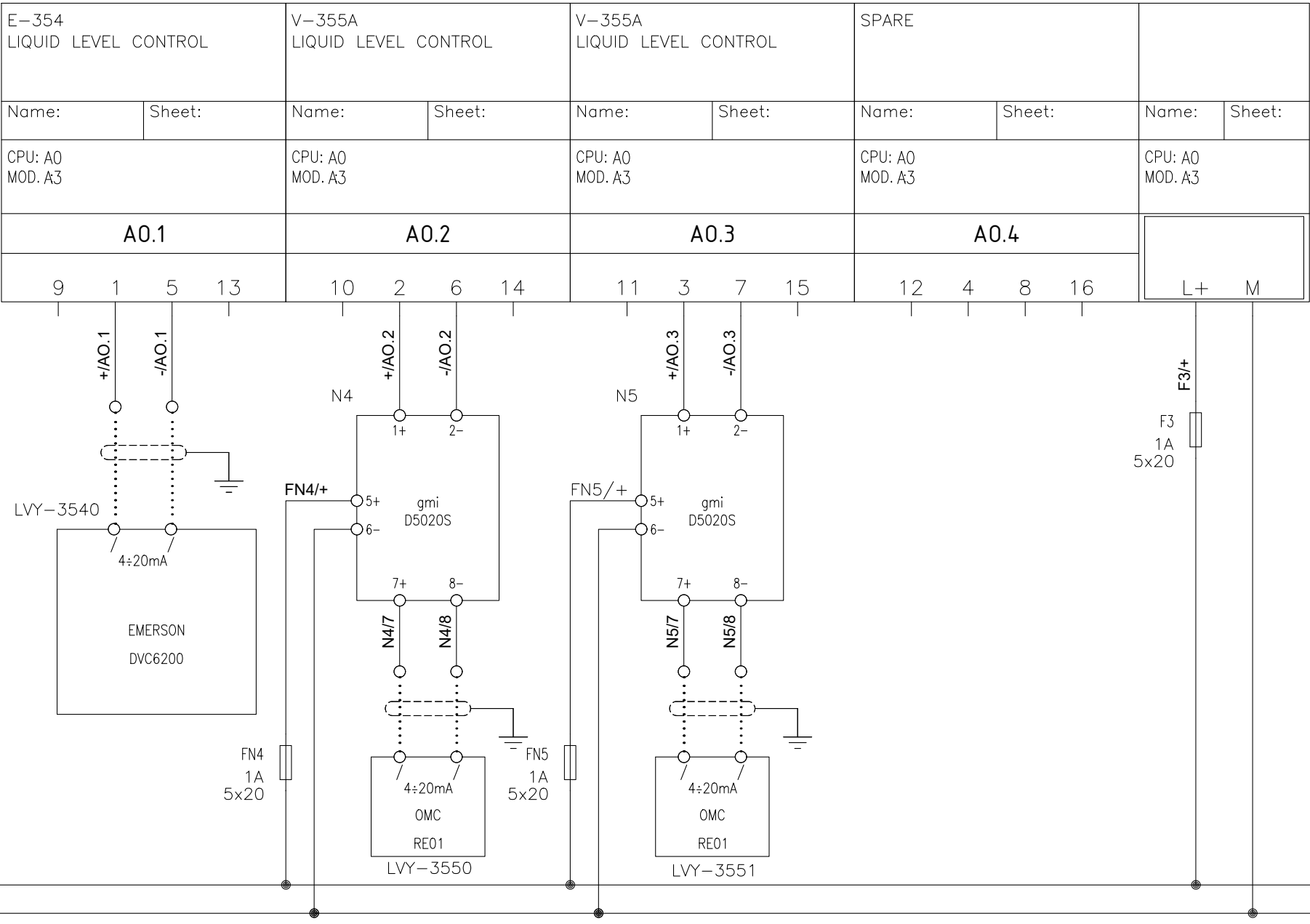


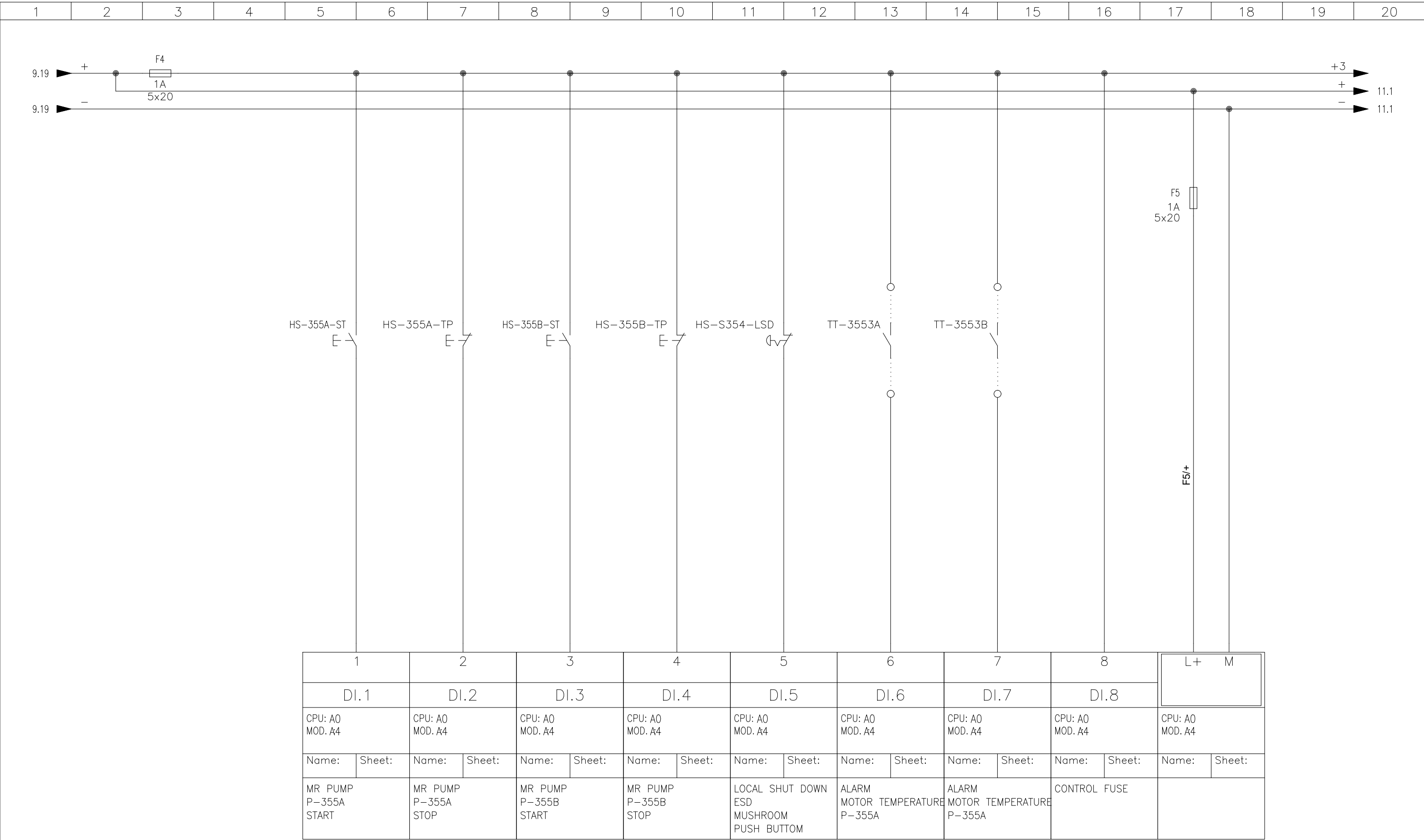
G1: TRIO-PS-2G/1AC/24DC/10 - Power supply unit PHORNIX CONTACT

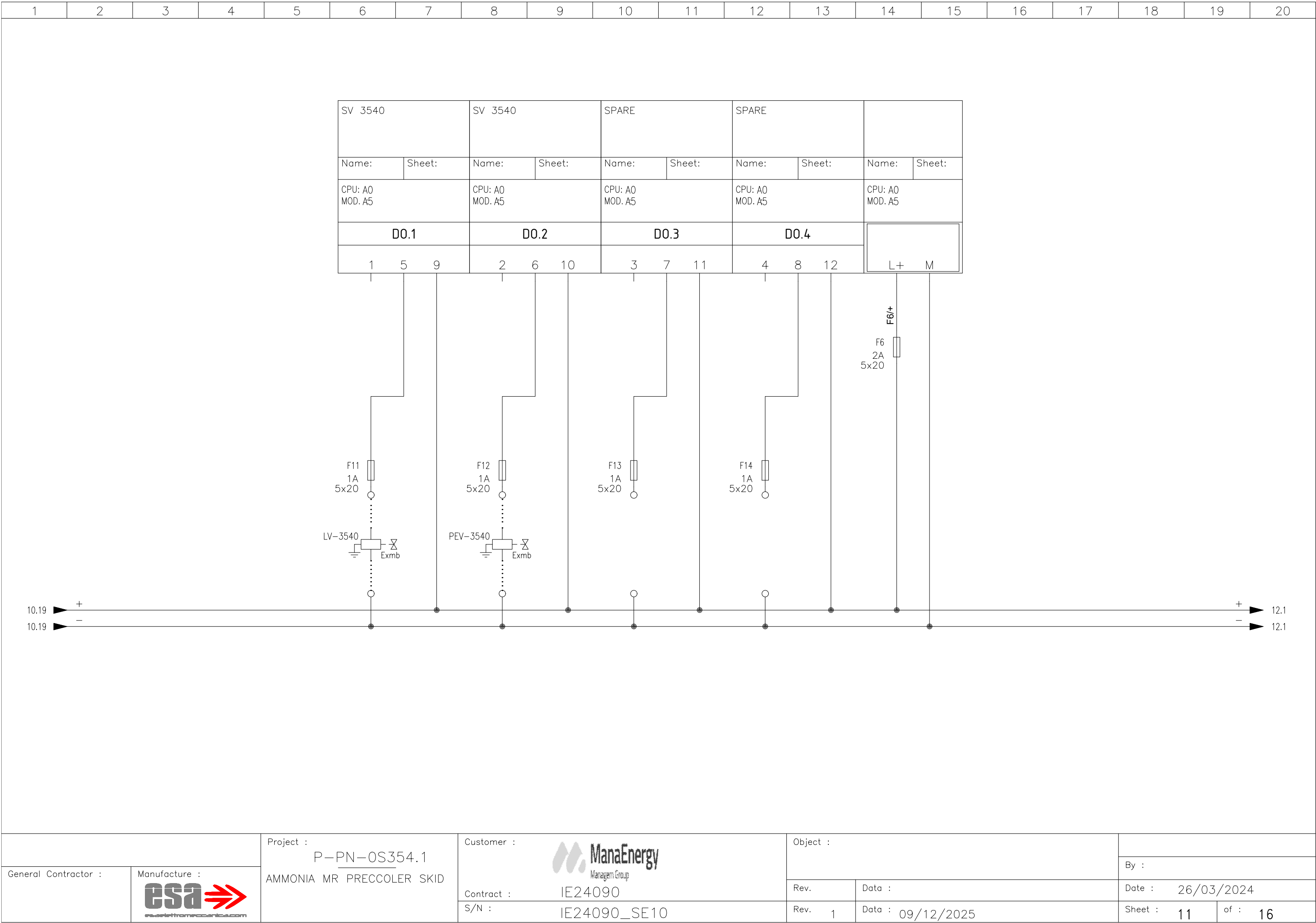
General Contractor : Manufacture : esa es.electrotecnica.com
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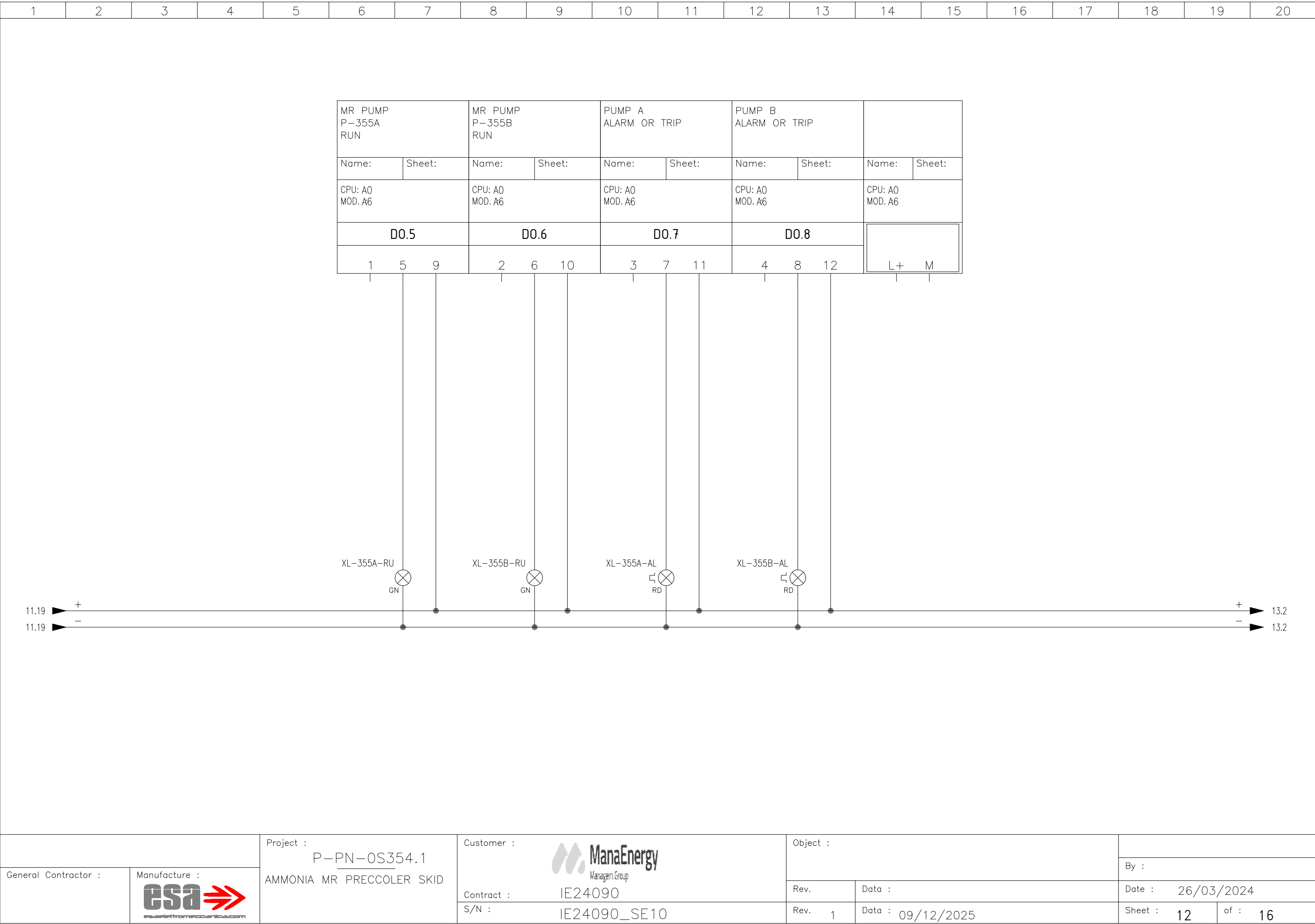




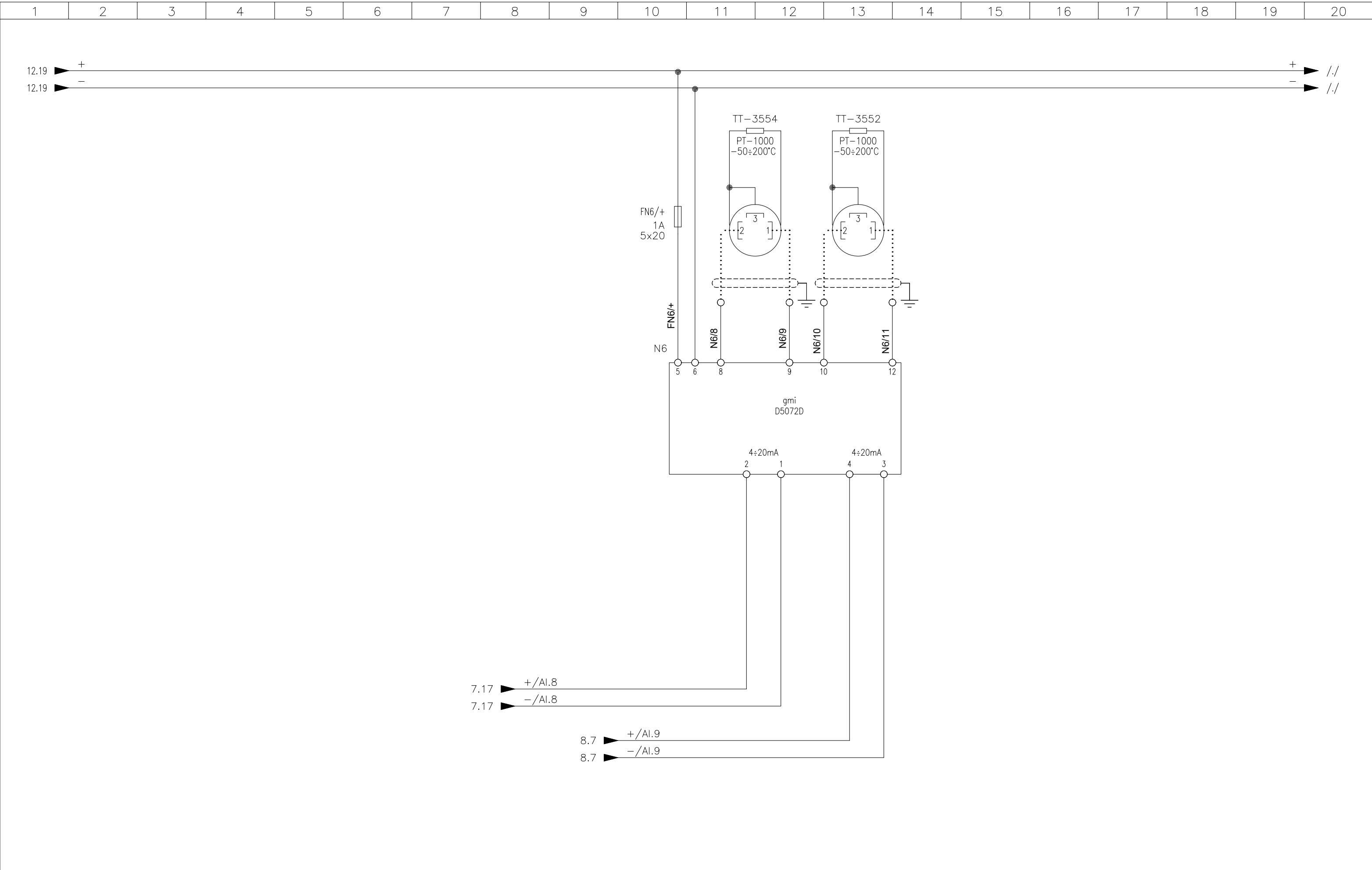


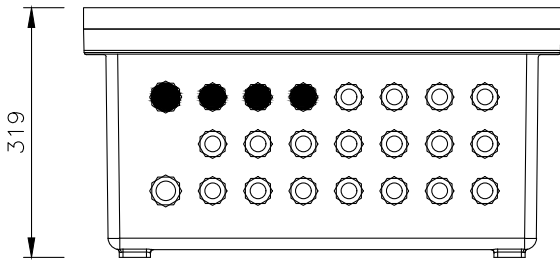
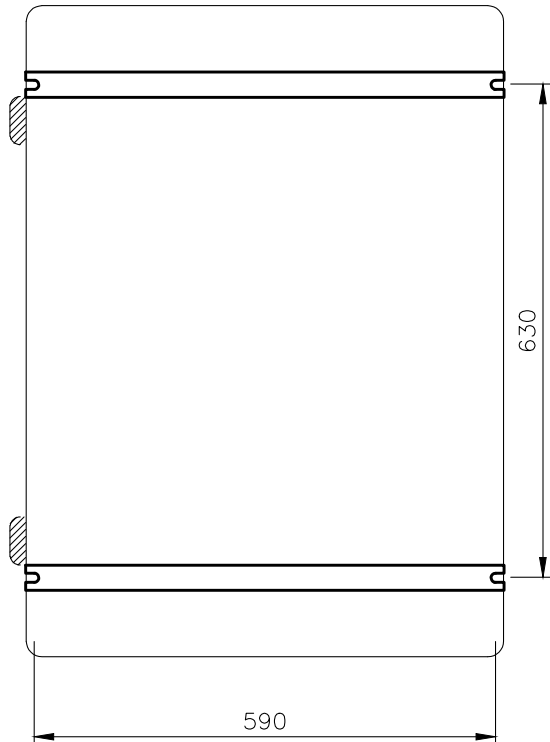
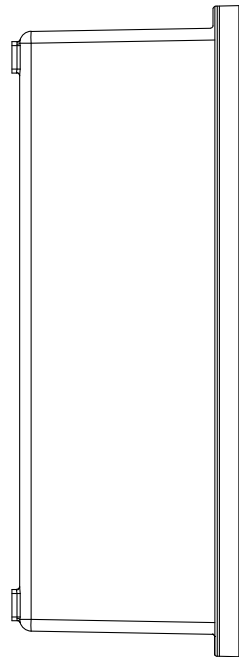
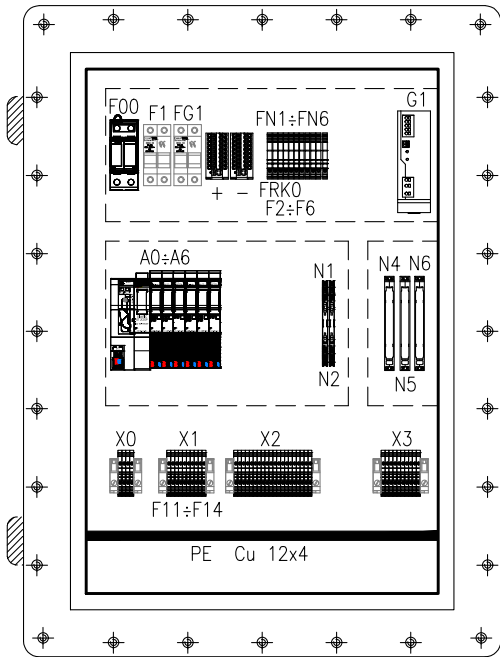
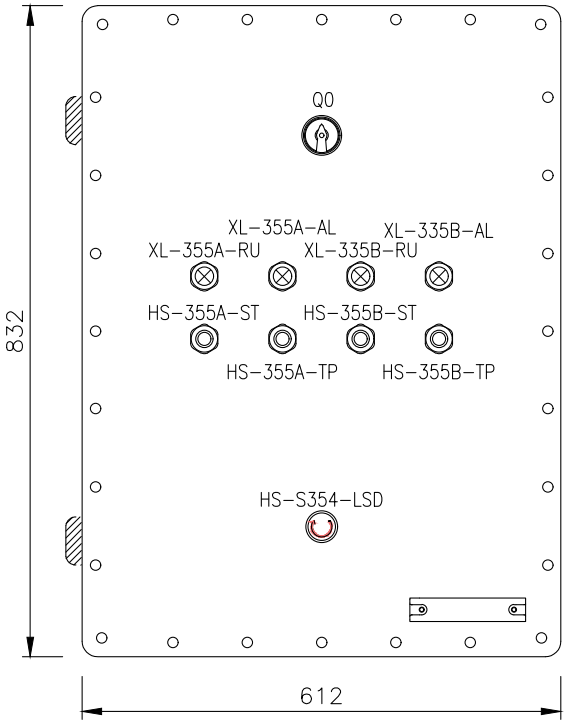






General Contractor :		Manufacture : 		Project : P-PN-0S354.1		Customer :  Contract : IE24090 S/N : IE24090_SE10				Object :					
				AMMONIA MR PRECCOLER SKID								By :			
										Rev. 1		Data : 09/12/2025		Date : 26/03/2024	
												Sheet : 12		of : 16	





MILANO-ITALY Tel 024455358 Fax 0248401741

Type: EJB-G
230Vac 50Hz 16A
Max Dis. P: 58 W

CE 0722 Ex II 2GD
INERIS18ATEX0026X/00
Ex db [ia Ga] IIB T6
Ex tb [ia Da] IIIC T85 °C IP66
-20°C < Tamb < +50°C

Cable entries: see instructions

WARNINGS: DO NOT OPEN WHILE ENERGIZED
DO NOT OPEN IF AN EXPLOSIVE ATMOSPHERE MAY BE PRESENT
READ INSTRUCTIONS NOTE BEFORE INSTALLATION AND USE

General Contractor : 		Project : P-PN-0S354.1 AMMONIA MR PRECCOLER SKID	Customer :  Contract : IE24090 S/N : IE24090_SE10		Object : Rev. 1 Data : 09/12/2025		By : Date : 26/03/2024	
							Sheet : 14	of : 16

INITIALS	SUPPLIER PN	DESCRIPTION	SUPPLIER
–	3273110	PTFIX 6/18X2,5–NS35 GY – Distribution block	PHOENIX CONTACT
+	3273110	PTFIX 6/18X2,5–NS35 GY – Distribution block	PHOENIX CONTACT
A0	6AG1155–6AU02–7BN0	ET 200SP, PROFINET interface module IM 155–6PN Standard	SIEMENS
A0	6AG1193–6AR00–7AA0	ET 200SP BA 2xRJ45	SIEMENS
A1	6AG1134–6GF00–7AA1	AI 8xl 2–/4–wire BA	SIEMENS
A1	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals	SIEMENS
A2	6AG1134–6GF00–7AA1	AI 8xl 2–/4–wire BA	SIEMENS
A2	6AG1193–6BP00–7BA0	ET 200SP BU15–P16+A0+2B push–in terminals	SIEMENS
A3	6AG1135–6HD00–7BA1	SIPLUS ET 200SP AQ 4xU/I Standard	SIEMENS
A3	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals	SIEMENS
A4	6AG1131–6BH01–7BA0	ET 200SP DI 16x24 V DC	SIEMENS
A4	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals	SIEMENS
A4	6AG1132–6GD51–7BA0	ET 200SP RQ 4x24VDC/2A CO ST	SIEMENS
A4	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals	SIEMENS
A5	6AG1132–6GD51–7BA0	ET 200SP RQ 4x24VDC/2A CO ST	SIEMENS
A5	6AG1193–6BP00–7BA0	ET 200SP BU15–P16+A0+2B push–in terminals	SIEMENS
F00	952110	DG M TT 2P 275 – SPD	DEHN
F1	2302038	Fuseholder for cylindrical fuses – 2P for fuses 10,3x38mm – BCH 2x38 32A 690V	ITALWEBER
F2	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F3	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F4	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F5	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F6	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FG1	2301138	Fuseholder for cylindrical fuses – 1P+N for fuses 10,3x38mm – BCH 1x38N 32A 690V	ITALWEBER
FN1	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN2	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN3	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN4	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN5	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN6	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FRK0	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
G1	2903149	TRIO–PS–2G/1AC/24DC/10 – Power supply unit	PHOENIX CONTACT
HS–355A–ST	PLA30	PUSH BUTTON GREEN	CMC
HS–355A–ST	E20GE11	CONTACT BLOCK N.O. +N.C. WITH FLANGE	ELFI
HS–355A–TP	PLA20	PUSH BUTTON RED	CMC
HS–355A–TP	E20GE11	CONTACT BLOCK N.O. +N.C. WITH FLANGE	ELFI
HS–355B–ST	PLA30	PUSH BUTTON GREEN	CMC
HS–355B–ST	E20GE11	CONTACT BLOCK N.O. +N.C. WITH FLANGE	ELFI
HS–355B–TP	PLA20	PUSH BUTTON RED	CMC
HS–355B–TP	E20GE11	CONTACT BLOCK N.O. +N.C. WITH FLANGE	ELFI
HS–S354–LSD	PLC–S	MUSHROOM HEAD MAINTAINED (TWIST TO RELEASE)	CMC
HS–S354–LSD	E20GE11	CONTACT BLOCK N.O. +N.C. WITH FLANGE	ELFI
N1	2902049	MINI MCR–2–RTD–UI – Resistance thermometer measuring transducer	PHOENIX CONTACT
N2	2902049	MINI MCR–2–RTD–UI – Resistance thermometer measuring transducer	PHOENIX CONTACT
N4	D5020S	I.S. SIL2 HART? Isolating Driver	GMI
N5	D5020S	I.S. SIL2 HART? Isolating Driver	GMI
N6	D5072D	I.S. SIL2 Multifunction Temperature Converter	GMI
P–PN–OS354.1	EJB–G	ENCLOSURE 612x832x319 mm L x H x D	CMC
Q0	PSRC–320	SELECTOR SWITCH	CMC
XL–355A–AL	PLD30	PILOT LAMP GREEN	CMC
XL–355A–AL	O20PD9	LAMPHOLDER BA9S, DIRECT SUPPLY	ELFI
XL–355A–RU	PLD20	PILOT LAMP RED	CMC
XL–355A–RU	O20PD9	LAMPHOLDER BA9S, DIRECT SUPPLY	ELFI
XL–355B–AL	PLD30	PILOT LAMP GREEN	CMC
XL–355B–AL	O20PD9	LAMPHOLDER BA9S, DIRECT SUPPLY	ELFI
XL–355B–RU	PLD20	PILOT LAMP RED	CMC
XL–355B–RU	O20PD9	LAMPHOLDER BA9S, DIRECT SUPPLY	ELFI

